MS200 Exam (May 8, 2020)

Instructions

You may use any written or printed materials. You may use a calculator. Please explain your answers clearly, write on only side of each sheet of paper, and number each sheet. If you use scratch paper, you do not need to upload it. You can take advantage of this to write more polished answers. You have 2.5 hours to complete this and 45 minutes to scan and upload your work.

QUESTION 1 [TOTAL MARKS: 13]

Consider for this question the four vectors

$$\mathbf{x} = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, \qquad \mathbf{y} = \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}, \qquad \mathbf{z} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \qquad \mathbf{w} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

(a) Compute $\mathbf{x} + 2\mathbf{y}$, $|\mathbf{y}|$, and the angle between $\mathbf{y}$ and $\mathbf{w}$.

[4 marks]

(b) The four vectors here are not linearly independent. There are numbers α, β, and γ such that $\alpha\mathbf{x} + \beta\mathbf{y} + \gamma\mathbf{z} = \mathbf{w}$. Find the matrix A such that

$$A \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \mathbf{w}.$$

[3 marks]

(c) Find the α, β, and γ introduced in part (b).

[6 marks]

QUESTION 2 [TOTAL MARKS: 9]

Consider the linear functions $\mathbf{f} : \mathbb{R}^2 \to \mathbb{R}^3$ and $g : \mathbb{R}^3 \to \mathbb{R}$ given by

$$\mathbf{f}(x_1, x_2) = \begin{pmatrix} x_1 \\ x_2 \\ x_1 - x_2 \end{pmatrix}, \qquad g(y_1, y_2, y_3) = y_1 + y_3 - 2y_2.$$

(a) Write down matrices A and B such that $\mathbf{f}(\mathbf{x}) = A\mathbf{x}$ and $g(\mathbf{y}) = B\mathbf{y}$, where $\mathbf{x}^T = (x_1 \quad x_2)$ and $\mathbf{y}^T = (y_1 \quad y_2 \quad y_3)$.

[3 marks]

(b) Say whether or not each of the following functions exist: $\mathbf{f} \circ g$, $g \circ \mathbf{f}$, $\mathbf{f} + 2g$.

[3 marks]

(c) Compute the matrix C for which $(g \circ \mathbf{f})(\mathbf{x}) = C\mathbf{x}$.

[3 marks]

QUESTION 3 [TOTAL MARKS: 19]

Let

$$A = \frac{1}{3}\begin{pmatrix} 2 & -1 & 2 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{pmatrix}, \qquad \mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

The matrix A is orthogonal and $\mathbf{v}$ is its only real eigenvector.

(a) Find $\mathbf{x}$ such that $A\mathbf{x} = 4\mathbf{v}$.

[Hint: This does not require a lot of computation.]

[5 marks]

(b) Find $\det A$.

[6 marks]

(c) If the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$ is $30°$, what is the angle between $A\mathbf{a}$ and $A\mathbf{b}$?

[3 marks]

(d) Find A^{-1}.

[5 marks]

QUESTION 4 [TOTAL MARKS: 10]

Consider a plane in $\mathbb{R}^3$ which intersects the origin and is orthogonal to

$$\mathbf{v} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}.$$

(a) Find a 3×3 matrix $P_{\mathbf{v}}$ which projects vectors onto this plane.

[5 marks]

(b) Given the matrix $P_{\mathbf{v}}$ from part (a), describe the geometric significance of the (nonlinear) function $f(\mathbf{w}) = \sqrt{\mathbf{w}^T P_{\mathbf{v}} \mathbf{w}}$.

[5 marks]

QUESTION 5 [TOTAL MARKS: 24]

Suppose that the components of a 4×4 matrix A are given by

$$(A)_{ij} = (-1)^{i+j} ij$$

for all $i, j \in \{1, \ldots, 4\}$.

(a) Is A square? Is it symmetric? Is it antisymmetric? Is it diagonal? Is it the identity matrix?

[5 marks]

(b) There is a vector $\mathbf{v}$ such that $A = \mathbf{v}\mathbf{v}^T$. Find $\mathbf{v}$.

[3 marks]

(c) It may be shown that $A^{n+1} = \alpha_n A^n$ for every positive integer n. Find the scalars α_n and comment on whether or not they depend on n.
[Hint: Part (b) should be helpful.]

[8 marks]

(d) Find four linearly-independent eigenvectors of A and their associated eigenvalues.

[Hint: Part (b) should be helpful.]

[8 marks]

QUESTION 6 [TOTAL MARKS: 9]

Consider the linear system

$$\begin{aligned} x_1 + 4x_2 + x_3 &= 1, \\ 3x_1 + x_2 &= -1, \end{aligned}$$

where x_1, x_2, and x_3 are unknowns.

(a) Write down the augmented matrix associated with this system.

[3 marks]

(b) Use Gauss-Jordan elimination to place your answer from part (a) into reduced row echelon form. Find all x_1, x_2, and x_3 which are solutions.

[6 marks]

QUESTION 7 [TOTAL MARKS: 16]

Suppose that a matrix A can be written as

$$A = P \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & -1/2 \end{pmatrix} P^{-1},$$

where

$$P = \begin{pmatrix} 1 & 2 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 3 \end{pmatrix}.$$

(a) Find all eigenvalues and eigenvectors of A. Make sure to indicate which eigenvalues are associated with which eigenvectors.

[5 marks]

(b) Suppose that a matrix B satisfies $B\mathbf{v} = 2\mathbf{v}$, where $\mathbf{v}$ is one of the eigenvectors of A. Compute $(AB - BA)\mathbf{v}$.

[6 marks]

(c) A^{10} can be written as PDP^{-1}, where D is a diagonal matrix. Find D.

[5 marks]